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Small deformation monitoring of Ultra-High Voltage transmission towers using China's high-resolution C-band SAR satellites

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Ultra-high voltage (UHV) transmission towers are critical infrastructures for the stability and resilience of power systems. Their structural integrity is constantly challenged by conductor tension, thermal expansion, and external loads. Conventional inspection techniques, including UAV surveys and in-situ sensors, are costly and spatially constrained. In contrast, spaceborne synthetic aperture radar (SAR) provides a cost-effective means for wide-area, millimeter-level deformation monitoring.

This study proposes a tower persistent scatterer (PS) simulation and interferometric elevation-phase modeling method that integrates three-dimensional LiDAR-derived tower models with the radar range-Doppler equation. Using high-resolution C-band SAR data from China's Fucheng-1 satellite, 78 ascending and descending scenes were analyzed over two 500 kV transmission lines in Chongqing. Corner reflectors (CRs) were installed at both tower bases and on tower bodies to provide accurate geometric calibration parameters and high-confidence CR-PS points for tower deformation analysis.

Results demonstrate that CR-based calibration achieved sub-pixel geometric accuracy and millimeter phase precision. The base CRs revealed approximately 8 mm of vertical subsidence over nine months. Tower-body CRs exhibited height-dependent small deformations corresponding to differential thermal expansion at different structural levels.

Two types of deformation estimation were performed: (1) Based on one-year short-baseline interferometric pairs, tower deformation ranges were empirically derived under various temperature intervals, indicating that straight towers exhibited larger deformation amplitudes than strain towers, with descending-track results exceeding 6 rad when $\Delta T > 25$ °C. (2) Using the proposed differential interferometric approach that removes simulated elevation phases,

continuous deformation patterns consistent with the empirical thresholds were retrieved, validating the physical effectiveness of the model.

In conclusion, this study confirms the feasibility of using China's high-resolution C-band SAR satellites for long-term, high-precision monitoring of UHV transmission tower deformation. The proposed methodology validates the capability of meter-resolution SAR systems to capture subtle structural deformations and provides a methodological foundation for assessing large-scale infrastructure responses to geological hazards, earthquakes, and typhoons in complex environments.